

Taras Sereda. ML Researcher & Engineer | [email](#) | [github](#) | [blog](#) | [linkedin](#)

Machine Learning Researcher and Engineer with years of expertise in audio, vision, and text processing. Developed state-of-the-art ASR, TTS, and computer vision systems. Built systems for autonomous GPU kernel generation and optimisation. Passionate about pushing the boundaries of human knowledge.

Research Interests: ML for systems, ML learning theory, simulated worlds.

Publications and patents

- [Speeding up PyTorch inference by 87% on Apple devices with AI-generated Metal kernels](#). Taras Sereda, Natalie Serrino, Zain Asgar. 2025
- [Pheme: Efficient and Conversational Speech Generation](#). Budzianowski Paweł, Sereda Taras, Cichy Tomasz, Vulić Ivan. 2024
- [Transcribe, Align and Segment: Creating speech datasets for low-resource languages](#). Taras Sereda. 2024
- [System and method for simultaneous multilingual dubbing of video-audio programs](#). Patent number: 11942093. Aleksandr Dubinsky, Taras Sereda. 2024

Talks

- **Kforge: Cross-Platform Kernel Generation and Optimization for AI Accelerators**. [EuroSys 2026 InSyDe workshop](#), 2026. Edinburgh, UK.
- [How Good is AI at Generating AI Kernels?](#) AI Benchmark Club, 2025. San Francisco, CA.
- [**VocalicsAI. Artificial intelligence removes language barriers**](#) FW days, 2021. Kyiv, Ukraine
- [**Waveglow. Generative modeling for audio synthesis**](#) FW days, 2019. Kyiv, Ukraine

Technical skills

- Programming: Python, C++, Rust, CUDA, Java
- ML/DL Frameworks: PyTorch, TensorFlow
- ML: self-supervised learning, CNNs, Transformers, MoEs.
- MLOps: Docker, Kubernetes, CI/CD, MLflow, Weights & Biases
- Cloud computing: Amazon AWS, Google GCP

Leadership skills

Maintains composure and sustained performance under pressure. Connects operational decisions to long-term objectives, anticipating challenges and aligning teams and resources to deliver meaningful outcomes. Makes sound, timely decisions in ambiguous environments where information is incomplete.

Professional Experience

Member of Technical Staff @ [Gimlet Labs](#)

San Francisco, CA (Nov 2024 - present)

- Leading research on LLM-powered kernel generation for GPU/accelerator optimization.
- Developing k-forge, an AI agent that autonomously generates and optimizes low-level kernels directly from PyTorch. Focusing on accelerations of both training and inference workloads across backends CUDA, ROCm, and Metal.

ML Consultant @ Self-employed

Sep 2022 - present

- Developed a real-time ASR system for low SNR environments (-10 to 0 dB), outperforming open-source SOTA models on domain-specific data
- Collaborated with PolyAI to create a zero-shot TTS system for call center automation, achieving faster-than-real-time performance on A100 GPUs

Visiting Scholar @ Ukrainian Catholic University

Lviv, Ukraine (Mar 2022 - present)

- Deliver lectures on speech and audio processing, focusing on TTS, ASR, and source separation(SS)
- Mentor and advise students on academic challenges in ML and audio processing
- Developed course material: github.com/taras-sereda/deep-learning-for-audio

Co-founder, Director of Research @ Vocalics.ai

Kyiv, Ukraine (Feb 2019 - Sep 2022)

- Led R&D for a novel multi-lingual speech synthesis system preserving speaker identity and style
- Managed a team of 4 researchers, driving innovation in SOTA speech synthesis technologies
- Implemented and customized multiple SOTA papers in speech synthesis, including autoregressive and parallel TTS approaches
- Developed methodology for quality evaluation of generated speech in dimensions of intelligibility, speaker and prosody similarity
- Conducted customer interviews and market research to validate product-market fit

ML Engineer @ Whisper.ai

San Francisco, CA, USA (Feb 2018 - Jul 2018)

- Developed advanced audio source separation models (PIT, Chimera network) deployed on ARM devices
- Improved listening experience for those with mild hearing loss in noisy environments

ML Researcher @ Ring Labs

Kyiv, Ukraine (Sep 2016 - Apr 2017)

- Established and led the ML department as the first employee in the Ukrainian R&D office
- Developed object detection algorithms based on R-CNN and YOLO with customized in-house implementations

ML Engineer (Part time) @ IPGraphy

Kyiv, Ukraine (Oct 2015 - Jun 2016)

- Developed core algorithm for visual object similarity search applied to trademark images
- Created an IP rights NN-based tool to enhance attorney productivity

ML Engineer @ DepositPhotos

Kyiv, Ukraine (Dec 2014 - Dec 2015)

Developed core algorithms for a virtual stylist app, including:

- Recommender system for clothing combination suggestions
- Neural networks for automatic clothes classification and wardrobe categorization

Education

Masters's Degree in Mathematical Modeling

*Kyiv, Ukraine, (2011 - 2012) KNEU

Bachelor's Degree in Cybernetics

Kyiv, Ukraine, (2007 - 2011) KNEU